

```
import cv2
import numpy as np
from google.colab import files
from google.colab.patches import cv2_imshow

# Upload image from your computer
uploaded = files.upload()

# Get the uploaded image name
filename = next(iter(uploaded))

# Read the image
image = cv2.imread(filename)

# Check if image is loaded
if image is None:
    print("Error: Could not load the image.")
else:
    # Laplacian filter kernel
    kernel = np.array([[0, 1, 0],
                       [1, -8, 1],
                       [0, 1, 0]])

    # Apply filter
    sharpened = cv2.filter2D(image, -1, kernel)

    # Display Original Image
    print("Original Image:")
    cv2_imshow(image)

    # Display Filtered Image
    print("Filtered Image:")
    cv2_imshow(sharpened)

    # Save output image
    cv2.imwrite("sharpened.jpg", sharpened)

    print("Output image saved as sharpened.jpg")
```

Choose Files sampled_image.jpg

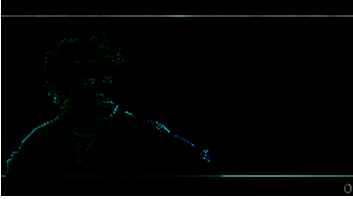
sampled_image.jpg(image/jpeg) - 9699 bytes, last modified: 7/22/2026 - 100% done

Saving sampled_image.jpg to sampled_image.jpg

Original Image:



Filtered Image:



Output image saved as sharpened.jpg